

From 673 failures to nine.

Reliability must hold across seeds.

On this selected benchmark, the mixed method uses reference labels during training and more total resources. Neither winner queries the reference at deployment. Every frozen policy is scored with no angle router and no mixture of checkpoints.

Task and return

Pendulum, 200 deterministic steps. $r = -(\theta^2 + 0.1\dot{\theta}^2 + 0.001a^2)$; R is the undiscounted sum. Torque a is in $[-2, 2]$.

Near reference

$$R_{\pi} \geq R^* - 5$$

R^* is the better stored dynamic-programming or controller return. The tolerance is five raw return units.

Task success

$\cos \theta \geq 0.95$ and $|\dot{\theta}| \leq 1$ rad/s for at least 80% of steps; no failure streak longer than 50 steps.

Strict win and grid

$R_{\pi} > R^*$ with no epsilon. $\theta = -\pi:\pi/30:\pi$; $\dot{\theta} = -1:0.05:1$ rad/s; actor seeds 0–4. Grid trials are correlated.

99.928%

RL + supervised near-reference success
12,496 / 12,505 trials

Task success 93.858% · strict wins 12.555%

94.618%

pure-RL near-reference success
11,832 / 12,505 trials

Task success 92.499% · strict wins 18.417%

1 · Method

Two clean routes to one frozen policy

RL + supervised

- TRAIN** SimbaV2 learns full-circle actions from stored dynamic-programming and controller references.
- TRAIN** DAgger lets the actor control the system, then labels the visited states with the reference action. Three rounds add 30k transitions. A 20k follow-up automatically prioritizes measured failures.
- DEPLOY** A shared 50k reward-only critic checks actor $\pm \{0, 0.05, 0.10\}$. FastSACn8 means a mostly one-step target plus a 0.775% eight-step auxiliary. Both critics must prefer a replacement.

Pure RL

- TRAIN** SimbaV2 SAC learns from reward for 100,000 transitions.
- DEPLOY** Project the actor through left-right symmetry: $a_{\text{sym}}(s) = [\pi(s) - \pi(Ms)] / 2$.
- DEPLOY** Search 41 torques in $[-2, 2]$. Replace a_{sym} only when both critics predict an improvement above 0.005.

Cost check · selected lineage per evaluated actor

Method	Learn	Discover	Labels
Automatic mixed	100k	800k	690k
Uniform mixed	100k	0	690k
Pure RL	100k	0	0

Mixed learning assigns 50k actor transitions plus one shared 50k critic. Uniform mixed removes discovery and still reaches **99.848%** near reference.

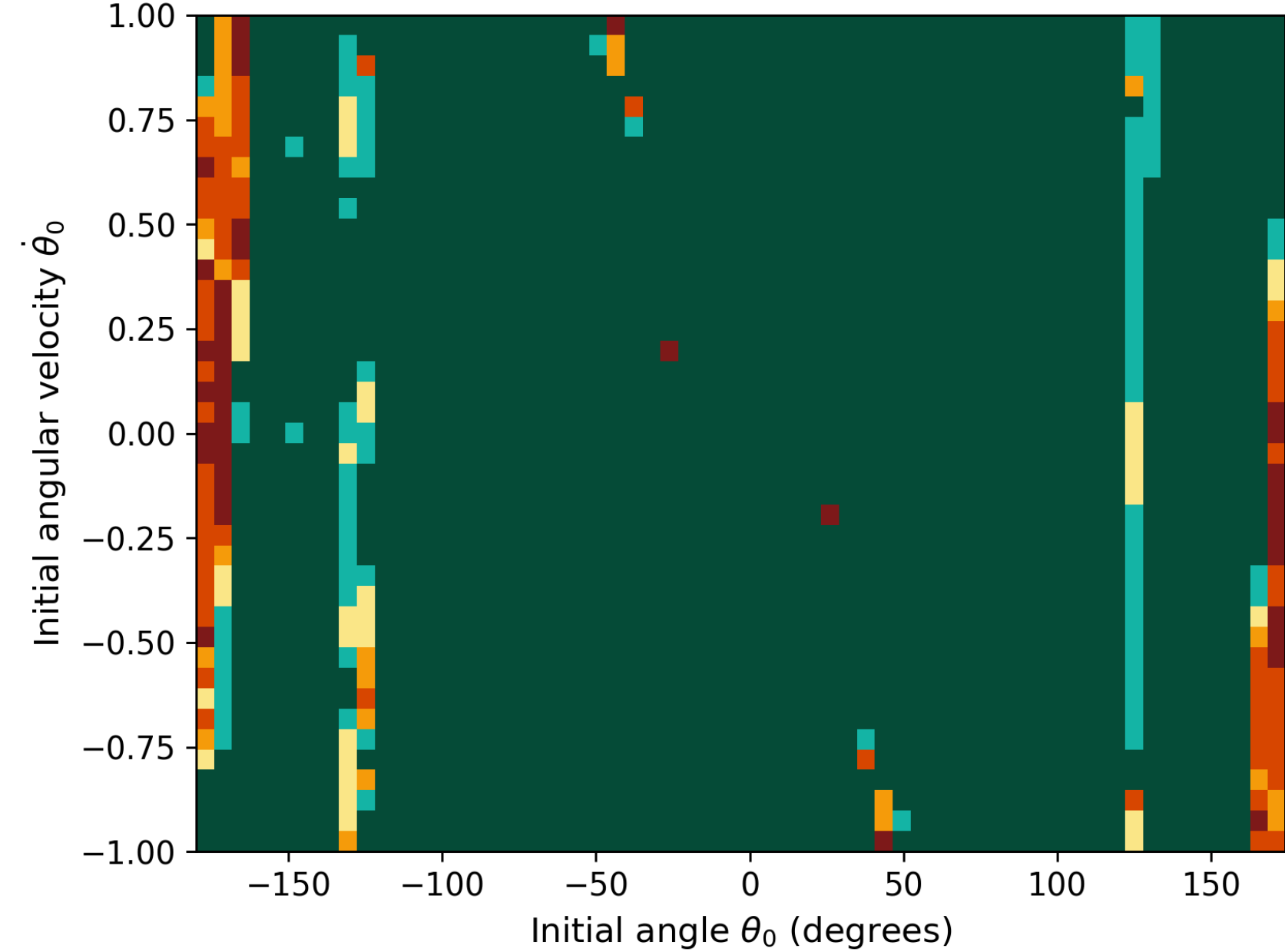
2 · Primary evidence

Nine failures versus 673

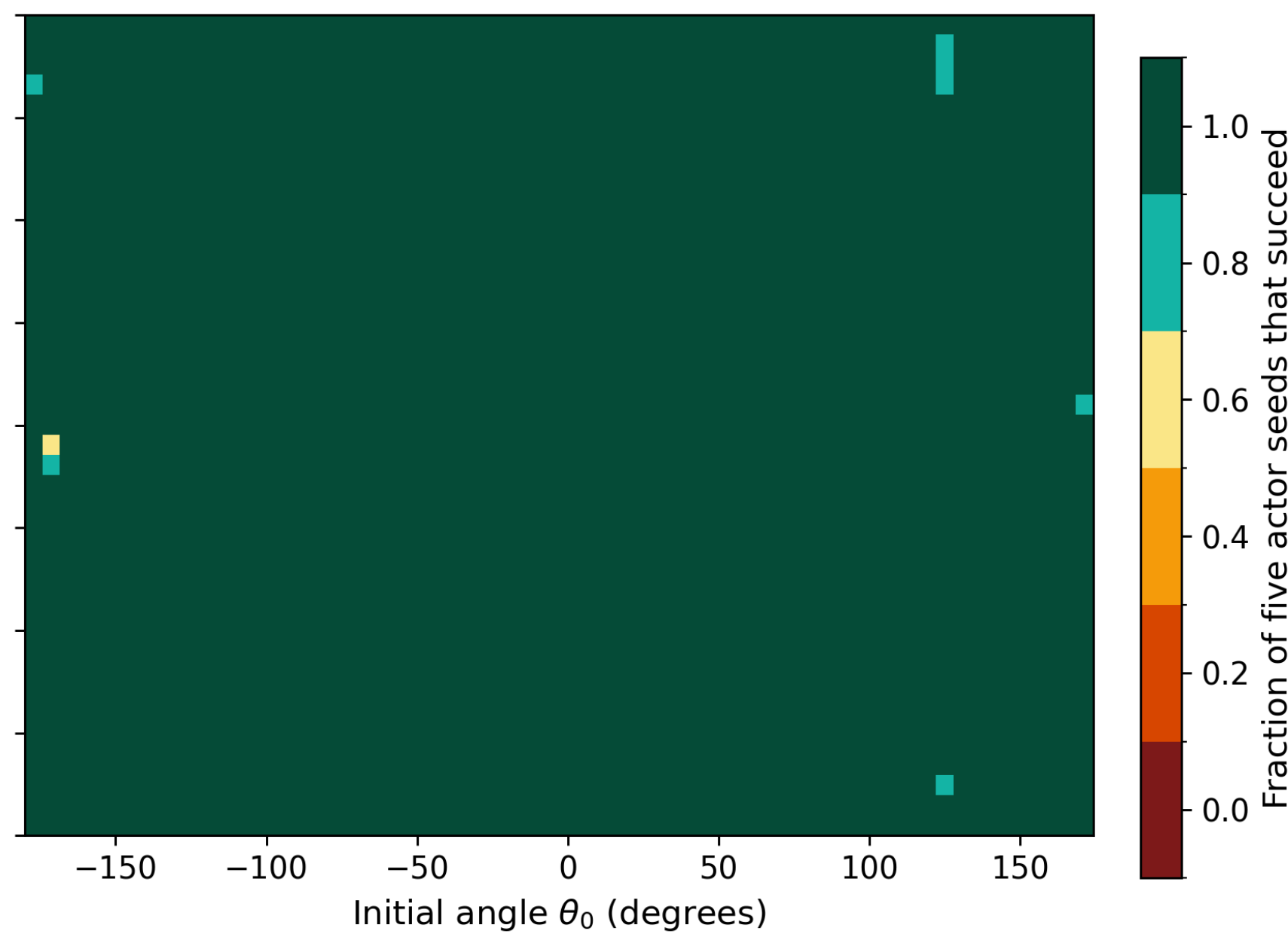
The grid map localizes failures. The tolerance curve checks whether the conclusion depends on choosing $\epsilon = 5$. Thin curves are individual actor seeds.

Where do failures occur?

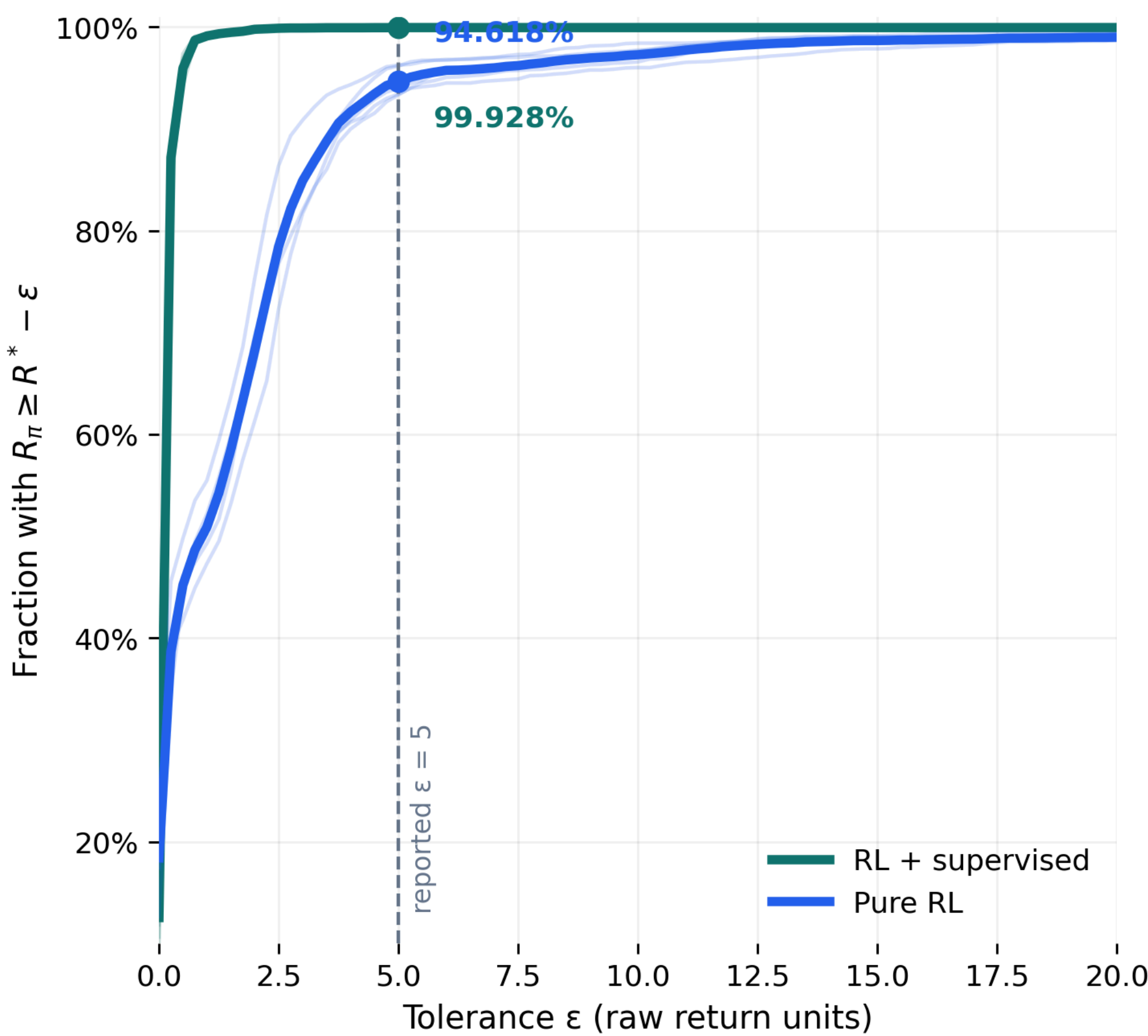
Pure RL: seed-dependent failure tail
673 failed seed-state trials



RL + supervision: nine sparse failures
9 failed seed-state trials



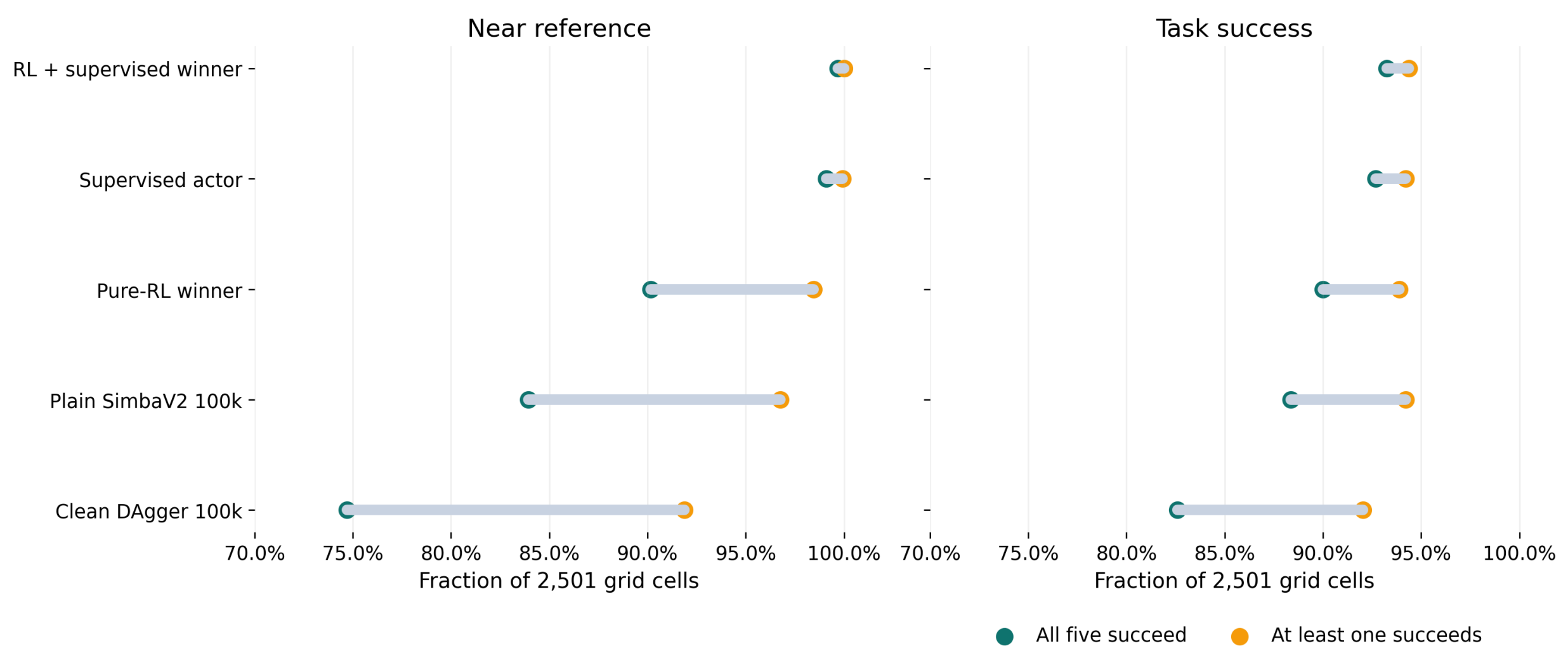
Does the tolerance choice change the result?



3 · Seed diagnosis

Pure RL can solve most cells, but not reliably

Near-reference range over actor seeds: mixed 99.720–100.000%; pure RL 93.323–96.242%.

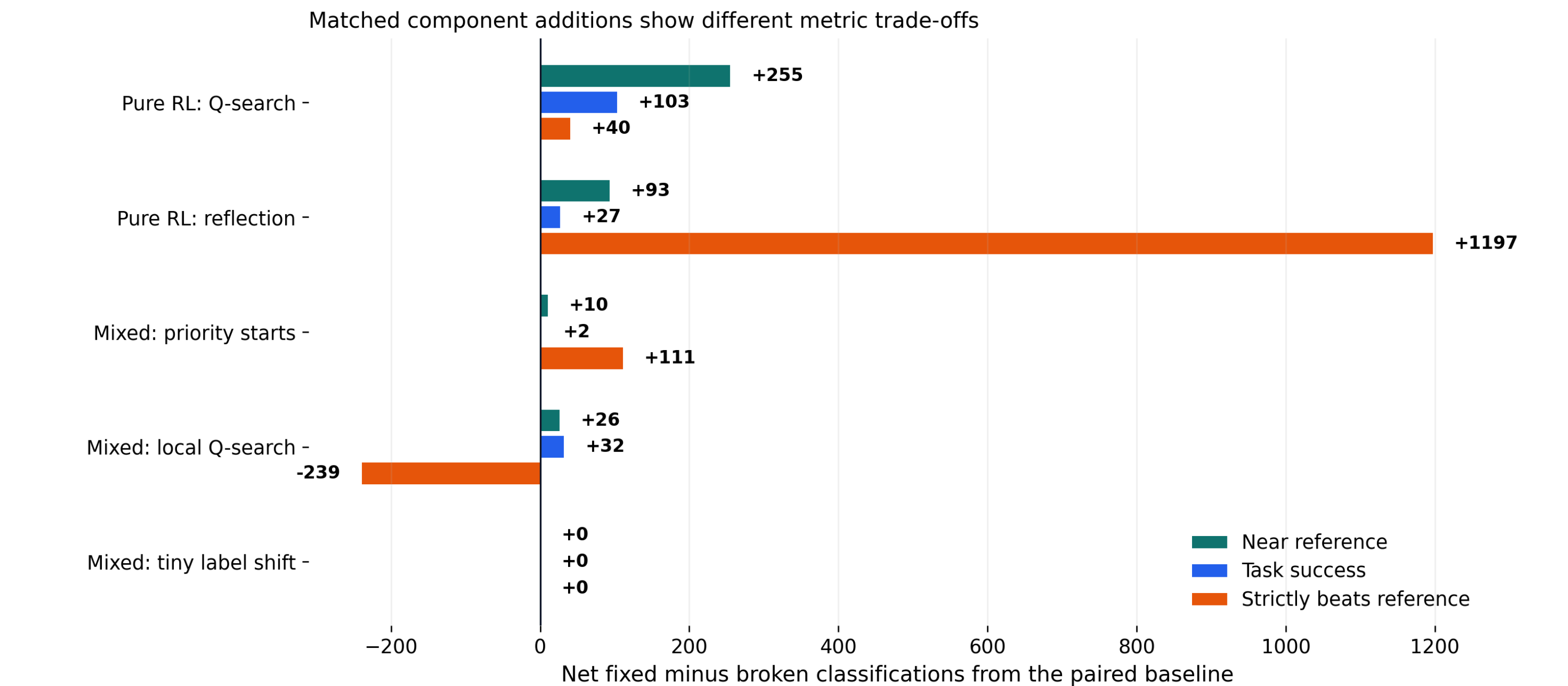


4 · Matched component checks

Which additions actually fix failures?

Each bar adds the named component to its paired baseline over 12,505 paired seed-state cases. Net count equals fixed minus newly broken classifications.

Near-reference rate: mixed actor 99.720% → 99.928% with local search; plain SimbaV2 91.835% → 94.618% with global search and reflection.



5 · Mechanism diagnostic

Critic uncertainty is larger than the local signal

Pairwise action ordering 84.4%

Harmful tested Q proposals 19.0%

Tail signal problem: median twin-critic disagreement is **2.43 times** the median local Q range. Typical rankings help, but harmful tested proposals occur 19.0% of the time on these targeted hard states.

Strict wins tell a different story. Pure RL beats the stored reference in 18.4% of trials, compared with 12.6% for the mixed method. Reliability and strict improvement are different operating points.

Definitions: pairwise ordering compares Q with realized counterfactual return. A harmful proposal lowers realized return. Diagnostics use five seeds, 99 hard states, and 21 actions per state.

Limits: one deterministic task, selected grid campaigns, and one shared mixed critic. The mechanism evidence is targeted and diagnostic, not causal or held out.

Take-home: mixed training suppresses seed-dependent failures. Pure RL often finds excellent actions, but its critic cannot gate tail corrections reliably enough.